Gianluca Levi - *Curriculum Vitae* April 23, 2026

Work Address 1	University of Iceland, Hjarðarhagi 2, 107 Reykjavík (Iceland)
Work Address 2	University of Trieste, Via Giorgieri 1, 34127 Trieste (Italy)
Born	June 17, 1990
Nationality	Italian
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Research profile and funding

Theoretical and experimental physical chemistry with a focus on spectroscopy and ultrafast dynamics. Development of density functional methods for modelling excited electronic states of molecules and solids. Atomic-scale simulations of photoinduced processes in the condensed phase using multiscale approaches. Time-resolved spectroscopy and X-ray scattering experiments at synchrotron and free electron laser facilities. Main areas of applications are solar energy conversion and photocatalysis.

Web page: https://gialev.github.io/
Google Scholar: https://scholar.google.com/
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Scopus Identifier: 57199649322

Research grants

2025 - *Simulations of electron transfer and atomic motion in photodriven molecular devices for enhanced solar energy conversion efficiency*

Type	Project grant
Funding source	Icelandic Research Fund
Role	Project Leader
Grant amount	606.078 Euros (including overhead)

2025 - NEXUS: *New excited state methods for overcoming challenges in sunlight conversion*

Type	ERC Starting Grant
Funding source	European Research Council
Role	Principal investigator
Grant amount	1.500.000 Euros

2021 - 2024 CNERGY: *Ultrafast Charge and Energy Transfer in Sunlight Conversion*

Type	Project grant
Funding source	Icelandic Research Fund
Role	Principal investigator
Grant amount	360.190 Euros (including overhead)

2019 - 2022 *Electron-Nuclear Dynamics in Iron-Sensitized Solar Cells: A Unified Model Description from Light Absorption to Charge Separation*

Type	Postdoctoral fellowship grant
Funding source	Icelandic Research Fund
Role	Principal investigator
Grant amount	224.240 Euros (including overhead)

2021 *Efficient Organic Solar Cells*

Type	Student project grant
Funding source	Student Innovation Fund, Icelandic Centre for Research
Role	Supervision of student project
Grant amount	4.080 Euros

Employment and education

Current and previous positions

2025 - present **Associate Professor**
Department of Chemical and Pharmaceutical Sciences, University of Trieste

2025 - present **Research Specialist**
Science Institute of the University of Iceland (Raunvísindastofnun Háskólans)

2018 - 2024 **Postdoctoral researcher**
Science Institute of the University of Iceland (Raunvísindastofnun Háskólans)

Education

2014 - 2018 **PhD studies**
Department of Chemistry, Technical University of Denmark (DTU)

Funding	Academic Excellence Scholarship
Thesis	*Photoinduced Molecular Dynamics in Solution: Multiscale Modelling and the Link to Ultrafast Experiments*
Supervisors	Prof. Klaus B. Møller, Prof. Niels E. Henriksen, Dr. Asmus O. Dohn

2012 - 2014 **Master of Science in Chemistry**
Polytechnic and Basic Sciences School, University of Naples Federico II

Final grade	110/110 & lode
Thesis	*Oxidation of solid carbon materials: structural and thermochemical analysis*
Supervisors	Prof. Mauro Causà, Dr. Ing. Osvalda Senneca, Prof. Piero Salatino

2009 - 2012 **Bachelor of Science in Chemistry**
Department of Chemistry, University of Naples Federico II

Final grade	110/110 & lode
Thesis	*Thermodynamic Study of the Metal-Water-Nicotine System*
Supervisor	Prof. Carla Manfredi

Other qualifications

2020 American Chemical Society Reviewer Lab, online course on peer review of scientific publications

Teaching and supervision

Teaching

2025 ***Programming for Computational Chemistry*** (6 ECTS)
Department of Chemical and Pharmaceutical Sciences, University of Trieste
Design of lectures, assignments and computer exercises in course for graduate students in chemistry

2022 - 2025 ***Molecular Spectroscopy and Reaction Dynamics*** (8/10 ECTS)
Physical Chemistry A (6 ECTS)
Computational Chemistry (8/10 ECTS)
Department of Chemistry, University of Iceland
Design of lectures, assignments and computer exercises in courses for graduate and undergraduate students in chemistry, biochemistry and chemical engineering

2023 ***Molecular Dynamics and X-ray Scattering Simulations*** (5 ECTS)
Department of Physics, Technical University of Denmark (DTU)
Design of computer exercises in special course for Master student in physics

2015 - 2016 ***Physical Chemistry 3*** (5 ECTS)
Applied Computational Chemistry (5 ECTS)
Department of Chemistry, Technical University of Denmark (DTU)

Teaching assistant in courses for graduate and undergraduate students in chemistry and chemical engineering

Supervision of students and postdoctoral fellows

	Postdocs
2026 - present	Dr. Ivan Tambovtsev, Icelandic Research Fund project, University of Iceland (co-supervisor)
2025 - present	Dr. Simon Petry, ERC NEXUS project, University of Trieste (supervisor)
2025 - present	Dr. Lorenzo Restaino, ERC NEXUS project, University of Iceland (supervisor)
2021 - 2023	Dr. Elli Selenius, Icelandic Research Fund CNERGY project, University of Iceland (supervisor)
	PhD students
2025 - present	Giulia Gamboni, ERC NEXUS project, University of Trieste (supervisor)
2025 - present	Diego L. Prieto, Icelandic Research Fund project, University of Iceland (supervisor)
2021 - 2024	Yorick L. A. Schmerwitz, Icelandic Research Fund PhD project, University of Iceland (co-supervisor)
2021 - 2024	Benedikt O. Birgisson, Icelandic Research Fund PhD project, University of Iceland (co-supervisor)
	Master students
2023	Magnus A. H. Christiansen, Technical University of Denmark (C´co-supervisor)
	Bachelor students
2021	Alec E. Sigurdarson, summer project funded by the Student Innovationn Fund of the Icelandic Centre for Research, University of Iceland (supervisor)

Research activity

Beamtimes awarded for experiments at large scale facilities

2022

Photoinduced intramolecular and solvation dynamics of an organic chromophore in polar solvent
Ultrafast X-ray solution scattering at SwissFEL X-ray free-electron laser (Paul Scherrer Institute)
Principal investigator, coordination and data analysis

Photoinduced intramolecular and solvation dynamics of an organic chromophore in polar solvent
Ultrafast X-ray solution scattering at European X-ray free electron laser
Main proposer, coordination and data analysis

Imaging photoinduced isomerization and solvation dynamics during intramolecular charge transfer in an organic chromophore via time-resolved X-ray scattering
Ultrafast X-ray solution scattering at Advanced Photon Source synchrotron (Argonne National Laboratory)
Principal investigator, coordination and data analysis

Publications

One monograph and 31 scientific articles (3 under review and currently only available as preprints on arXiv or Research Square). Six article as last author (one shared last authorship), eight as first author, ten as corresponding author, and two Editors' suggestions (*J. Chem. Phys.* 2023 and *Phys. Rev. Lett.* 2019). Total number of citations: 1141 (Google Scholar), 823 (Web of Science), 814 (Scopus). H-index: 18 (Google Scholar), 16 (Web of Science), 16 (Scopus).

Reviewing activity

Peer review for American Chemical Society journals *J. Chem. Theory Comput.*, *J. Phys. Chem. Lett.* and *J. Org. Chem.*, and Royal Society of Chemistry journal *Phys. Chem. Chem. Phys.*

Contributions at conferences and seminars (selection)

8 invited and 8 selected talks at international conferences. 5 poster contributions. 15 academic seminars.

2025 Laboratoire de Chimie Quantique (LCQ), Strasbourg University
Seminar

Condensed Matter Seminar Series, Scuola Internazionale Superiore di Studi Avanzati (SISSA)
Seminar

Symposium on electronic structure calculations, University of Iceland
Seminar

Physical and Theoretical Chemistry Laboratory, University of Oxford
Seminar

2024 Center for Interstellar Catalysis, InterCat
Online seminar

GPAW 2024 developer meeting, online
Invited talk

Department of Chemical and Pharmaceutical Sciences, University of Trieste
Seminar

Department of Molecular Sciences and Nanosystems, Ca' Foscari University of Venice
Seminar

Department of Applied Science and Technology, Polytechnic University of Turin
Seminar

20th International Conference on Density Functional Theory and its Applications (DFT 2024), École Nationale Supérieure de Chimie (Paris)
Selected talk

Progress in Ensemble Density Functional Theory: Opportunities and Challenges (EDFT workshop), Durham University
Invited talk

DanScatt XFEL workshop, University of Aarhus
Invited talk

20th ETSF Young Researchers' Meeting, Université Paul Sabatier (Toulouse)
Invited talk

Symposium on Excitations in Complex Systems: Recent Progress in the Ab Initio and Many-Body Theory Communities, American Physical Society March meeting 2024, Minneapolis
Invited talk

2023 Icelandic research e-infrastructure (IREI) conference, University of Iceland
Invited talk

15th Femtochemistry Conference, Freie Universität Berlin
Selected talk

Workshop on Emerging Excited State Methods in Electronic Structure, Toulouse
Invited talk

Laboratoire de Chimie et Physique Quantiques (LCPQ), Université Paul Sabatier (Toulouse)
Seminar

2022 Symposium on New Strategies for Solving the Kohn-Sham Equations, Conference on Scientific Computation and Differential Equations (SciCADE), University of Iceland
Selected talk

Symposium on Developments in Hybrid QM/QM, QM/MM, and Fragmentation Methods, American Chemical Society March meeting 2022, San Diego
Selected talk

Symposium on Density Functional Theory and Beyond, American Physical Society March meeting 2022, Chicago
Selected talk

2021 Chemistry-Biochemistry seminar series, University of Iceland

Online seminar

Quantum Molecular Dynamics (QMD) seminar series, University of Edinburgh
Online seminar

GPAW 2021: Users and Developers Meeting, online
Invited talk

Quantum Dynamics and Spectroscopy of Functional Molecular Materials and Biological Photosystems, Les Houches workshop 2021, online
Selected talk

2020 Computational Atomic-scale Materials Design (CAMD), Department of Physics, Technical University of Denmark (DTU)
Online seminar

New horizons in density functional theory Faraday Discussion conference, online
Selected talk

2019 Young Academics of Háskóli Ísland (YAHI) seminar series, University of Iceland
Seminar

2018 Department of Materials Science, University of Milano-Bicocca
Seminar

Chemistry-Biochemistry seminar series, University of Iceland
Seminar

2017 22nd International Symposium on Photochemistry and Photophysics of Coordination Compounds (ISPPCC 2017), Oxford
Selected talk

Organization of scientific meetings

2026 Main organizer of the 16th Nordic Baltic Femtochemistry Conference (NBFC26), University of Iceland
https://theochemui.github.io/nbfc26/

2022 Organizer of symposium on New Strategies for Solving the Kohn-Sham Equations, Conference on Scientific Computation and Differential Equations (SciCADE), University of Iceland

Code development in open-source software

Developer of the GPAW program for electronic structure calculations and active contributor to the Atomic Simulation Environment (ASE) package for atomic calculations. All developed code is freely available on gitlab, either on personal (https://gitlab.com/glevi, https://gitlab.com/theochem-ui) or official (https://gitlab.com/gpaw/gpaw, https://gitlab.com/ase/ase) repositories.

Awards

- Springer Thesis Award 2019. Monetary award and publication of the PhD thesis in the series *Springer Theses: Recognizing Outstanding Ph.D. Research* (10.1007/978-3-030-28611-8).
- *Premio Università Paolo Iannotti 2015 Laureati Eccellenti* awarded by the Ateneapoli magazine to students in Universities in Campania (Italy) who excel in their academic studies.

List of publications

(1) Haubro, M. L.; Zederkof, D. B.; Khakhulin, D.; Yousef, H.; Jiang, Y.; Bermudez, I. J. B.; Kluyver, T.; Lima, F.; Biednov, M.; Milne, C.; Uemura, Y.; Zalden, P.; Wang, H.; Dohn, A. O.; Mitterer, K.; Markmann, V.; Ohlson, O.; Wachter-Lehn, M. T.; Lenzen, P.; Jensen, B. K.; Høholdt, A.; van Driel, T. B.; Nimmrich, A.; Selenius, E.; **Levi, G.**; Møller, B.; Nielsen, M. M.; Haldrup, K. Analysis strategies for time-resolved X-ray solution scattering at high repetition rate XFEL sources. *Journal of Synchrotron Radiation* **2026**, *33*, 18–31, DOI: 10.1107/S1600577525009737.

(2) Mitterer, K. M.; Selenius, E.; Haubro, M. L.; Christiansen, M. A. H.; Markmann, V.; Hansen, B. L.; Krell-Jørgensen, M.; Hoock, J. G. F.; Lorentzen, V.; Beale, E. V.; Johnson, P. J. M.; Gosztola, D. J.; Cirelli, C.; Bacellar, C.; Dohn, A. O.; Laraia, L.; Møller, K. B.; Haldrup, K.; **Levi, G.**; Nielsen, M. M. Atomic-Scale Observation of Polar Solvation Dynamics Triggered by Photoinduced Charge Transfer in an Organic Chromophore. *Research Square* **2025**, *Shared last authorship*, DOI: 10.21203/rs.3.rs-7775086/v1.

(3) **Levi, G.**; Kroesbergen, M.; Thirion, L.; Schmerwitz, L. A. Y.; Jónsson, E. Ö.; Bilous, P.; Hansmann, P.; Jónsson, H. Orbital Optimization and Neural-Network-Assisted Configuration Interaction Calculations of Rydberg States. *arXiv:2510.26751* **2025**, DOI: 10.48550/arXiv.2510.26751.

(4) Thirion, L.; Schmerwitz, Y. L.; Kroesbergen, M.; **Levi, G.**; Jónsson, E. Ö.; Bilous, P.; Jónsson, H.; Hansmann, P. Natural-Orbital-Based Neural Network Configuration Interaction. *arXiv:2510.27665* **2025**, DOI: 10.48550/arXiv.2510.27665.

(5) Schmerwitz, Y. L. A.; Selenius, E.; **Levi, G.** Freeze-and-release direct optimization method for variational calculations of excited electronic states. *arXiv:2501.18568* **2025**, DOI: 10.48550/arXiv.2501.18568.

(6) Tambovtsev, I.; Schmerwitz, Y. L. A.; **Levi, G.**; Darmoroz, D. D.; Nesterov, P. V.; Orlova, T.; Jónsson, H. Fine Tuning of the Rotational Rate of Light-Driven, Second Generation Molecular Motors by Fluorine Substitutions. *The Journal of Physical Chemistry Letters* **2025**, *16*, 4014–4020, DOI: 10.1021/acs.jpclett.5c00408.

(7) Schmerwitz, Y. L. A.; Thirion, L.; **Levi, G.**; Jónsson, E. Ö.; Bilous, P.; Jónsson, H.; Hansmann, P. A Neural-Network-Based Selective Configuration Interaction Approach to Molecular Electronic Structure Yorick. *Journal of Chemical Theory and Computation* **2025**, *21*, 2301–2310, DOI: 10.1021/acs.jctc.4c01479.

(8) Birgisson, B. O.; Dohn, A. O.; Jónsson, H.; **Levi, G.** Decoherence and vibrational energy relaxation of the electronically excited PtPOP complex in solution. *Journal of Chemical Physics* **2025**, *162*, 044306, DOI: 10.1063/5.0241573.

(9) Selenius, E.; Sigurdarson, A. E.; Schmerwitz, Y. L. A.; **Levi, G.** Orbital optimized vs time-dependent density functional calculations of intramolecular charge transfer excited states. *Journal of Chemical Theory and Computation* **2024**, *20*, 3809–3822, DOI: 10.1021/acs.jctc.3c01319.

(10) Schmerwitz, Y. L. A.; Ollé, N. U.; **Levi, G.**; Jónsson, H. Saddle Point Search Algorithms for Variational Density Functional Calculations of Excited Electronic States with Self-Interaction Correction. *Proceedings of the Platform for Advanced Scientific Computing Conference (PASC '24)* **2024**, 1–11, DOI: 10.1145/3659914.3659933.

(11) Powers-Riggs, N. E.; Birgisson, B. O.; Raj, S. L.; Lenzen, P.; Zederkof, D. B.; Haubro, M.; Tveiten, K. V. D.; Hartsock, R. W.; Driel, T. B. V.; Chollet, M.; Robinson, J. S.; Nelson, S.; Haldrup, K.; Pedersen, K. S.; **Levi, G.**; Dohn, O.; Hannes, J.; Møller, K. B.; Natan, A.; Nielsen, M. M.; Gaffney, K. Characterization of deformational isomerization potential and interconversion dynamics with ultrafast X-ray solution scattering. *Journal of the American Chemical Society* **2024**, *146*, 13962–13973, DOI: 10.1021/jacs.4c00817.

(12) Mortensen, J. J.; Larsen, A. H.; Kuisma, M.; Ivanov, A. V.; Taghizadeh, A.; Peterson, A.; Haldar, A.; Dohn, A. O.; Schäfer, C.; Jónsson, E. Ö.; Hermes, E. D.; Nilsson, F. A.; Kastlunger, G.; **Levi, G.**; Jónsson, H.; Häkkinen, H.; Fojt, J.; Kangsabanik, J.; Sødequist, J.; Lehtomäki, J.; Heske, J.; Enkovaara, J.; Winther, K. T.; Dulak, M.; Melander, M. M.; Ovesen, M.; Louhivuori, M.; Walter, M.; Gjerding, M.; Lopez-Acevedo, O.; Erhart, P.; Warmbier, R.; Würdemann, R.; Kaappa, S.; Latini, S.; Boland, T. M.; Bligaard, T.; Skovhus, T.; Susi, T.; Maxson, T.; Rossi, T.; Chen, X.; Schmerwitz, Y. L. A.; Schiøtz, J.; Olsen, T.; Jacobsen, K. W.; Thygesen, K. S. GPAW: An open Python package for electronic structure calculations. *Journal of Chemical Physics* **2024**, *160*, 092503, DOI: 10.1063/5.0182685.

(13) Sigurdarson, A. E.; Schmerwitz, Y. L. A.; Tveiten, D. K. V.; **Levi, G.**; Jónsson, H. Orbital-optimized Density Functional Calculations of Molecular Rydberg Excited States with Real Space Grid Representation and Self-Interaction Correction (*Editor's Pick*). *Journal of Chemical Physics* **2023**, *159*, 214109, DOI: 10.1063/5.0179271.

(14) Katayama, T.; Choi, T. K.; Khakhulin, D.; Dohn, A. O.; Milne, C. J.; Vankó, G.; Németh, Z.; Lima, F. A.; Szlachetko, J.; Sato, T.; Nozawa, S.; Adachi, S. I.; Yabashi, M.; Penfold, T. J.; Gawelda, W.; **Levi, G.** Atomic-scale observation of solvent reorganization influencing photoinduced structural dynamics in a copper complex photosensitizer. *Chemical Science* **2023**, *14*, 2572–2584, DOI: 10.1039/d2sc06600a.

(15) Ivanov, A. V.; Schmerwitz, Y. L. A.; **Levi, G.**; Jónsson, H. Electronic excitations of the charged nitrogen-vacancy center in diamond obtained using time-independent variational density functional calculations. *SciPost Phys.* **2023**, *15*, 009, DOI: 10.21468/SciPostPhys.15.1.009.

(16) Schmerwitz, Y. L. A.; **Levi, G.**; Jónsson, H. Calculations of Excited Electronic States by Converging on Saddle Points Using Generalized Mode Following. *Journal of Chemical Theory and Computation* **2023**, *19*, 3634–3651, DOI: 10.1021/acs.jctc.3c00178.

(17) Schmerwitz, Y. L. A.; Ivanov, A. V.; Jónsson, E.; Jónsson, H.; **Levi, G.** Variational Density Functional Calculations of Excited States: Conical Intersection and Avoided Crossing in Ethylene Bond Twisting. *Journal of Physical Chemistry Letters* **2022**, *13*, 3990–3999, DOI: 10.1021/acs.jpclett.2c00741.

(18) Ivanov, A. V.; **Levi, G.**; Jónsson, E. Ö.; Jónsson, H. Method for Calculating Excited Electronic States Using Density Functionals and Direct Orbital Optimization with Real Space Grid or Plane-Wave Basis Set. *Journal of Chemical Theory and Computation* **2021**, *17*, 5034–5049, DOI: 10.1021/acs.jctc.1c00157.

(19) **Levi, G.**; Ivanov, A. V.; Jónsson, H. Variational Density Functional Calculations of Excited States via Direct Optimization. *Journal of Chemical Theory and Computation* **2020**, *16*, 6968–6982, DOI: 10.1021/acs.jctc.0c00597.

(20) **Levi, G.**; Ivanov, A. V.; Jonsson, H. Variational Calculations of Excited States Via Direct Optimization of Orbitals in DFT. *Faraday Discussions* **2020**, *224*, 448, DOI: 10.1039/D0FD00064G.

(21) Brandenburg, J. G.; Burke, K.; Fromager, E.; Gatti, M.; Giarrusso, S.; Gidopoulos, N. I.; Gori-Giorgi, P.; Gowland, D.; Helgaker, T.; Hodgson, M. J. P.; Lacombe, L.; **Levi, G.**; Loos, P.-F.; Maitra, N. T.; Maurina Morais, E.; Mehta, N.; Monti, F.; Mulay, M. R.; Pernal, K.; Reining, L.; Romaniello, P.; Ryder, M. R.; Savin, A.; Sirbu, D.; Teale, A. M.; Thom, A. J. W.; Truhlar, D. G.; Wetherell, J.; Yang, W. New approaches to study excited states in density functional theory: general discussion. *Faraday Discussions* **2020**, *224*, 483–508, DOI: 10.1039/d0fd90026e.

(22) **Levi, G.**; Causà, M.; Cortese, L.; Salatino, P.; Senneca, O. On how mild oxidation affects the structure of carbons: Comparative analysis by different techniques. *Applications in Energy and Combustion Science* **2020**, *1-4*, 100006, DOI: 10.1016/j.jaecs.2020.100006.

(23) **Levi, G.**; Biasin, E.; Dohn, A. O.; Jónsson, H. On the interplay of solvent and conformational effects in simulated excited-state dynamics of a copper phenanthroline photosensitizer. *Physical Chemistry Chemical Physics* **2020**, *22*, 748–757, DOI: 10.1039/C9CP06086C.

(24) Haldrup, K.; **Levi, G.**; Biasin, E.; Vester, P.; Laursen, M. G.; Beyer, F.; Kjær, K. S.; Van Driel, T. B.; Harlang, T.; Dohn, A. O.; Hartsock, R. J.; Nelson, S.; Glownia, J. M.; Lemke, H. T.; Gaffney, K. J.; Henriksen, N. E.; Møller, K. B.; Nielsen, M. M. Ultrafast X-Ray Scattering Measurements of Coherent Structural Dynamics on the Ground-State Potential Energy Surface of a Diplatinum Molecule (*Editors' Suggestion*). *Physical Review Letters* **2019**, *122*, 063001, DOI: 10.1103/PhysRevLett.122.063001.

(25) Abedi, M.; **Levi, G.**; Zederkof, D. B.; Henriksen, N. E.; Pápai, M.; Møller, K. B. Excited-state solvation structure of transition metal complexes from molecular dynamics simulations and assessment of partial atomic charge methods. *Phys. Chem. Chem. Phys.* **2019**, *21*, 4082–4095, DOI: 10.1039/C8CP06567E.

(26) Pápai, M.; Abedi, M.; **Levi, G.**; Biasin, E.; Nielsen, M. M.; Møller, K. B. Theoretical Evidence of Solvent-Mediated Excited-State Dynamics in a Functionalized Iron. *The Journal of Physical Chemistry C* **2019**, DOI: 10.1021/acs.jpcc.8b10768.

(27) **Levi, G.**, *Photoinduced Molecular Dynamics in Solution: Multiscale Modelling and the Link to Ultrafast Experiments, Multiscale Modelling and the Link to Ultrafast Experiments*; Springer Theses: Recognizing Outstanding Ph.D. Research; Springer: **2019**, DOI: 10.1007/978-3-030-28611-8.

(28) **Levi, G.**; Pápai, M.; Henriksen, N. E.; Dohn, A. O.; Møller, K. B. Solution structure and ultrafast vibrational relaxation of the PtPOP complex revealed by ΔSCF-QM/MM Direct Dynamics simulations. *Journal of Physical Chemistry C* **2018**, *122*, 7100–7119, DOI: 10.1021/acs.jpcc.8b00301.

(29) Biasin, E.; van Driel, T. B.; **Levi, G.**; Laursen, M. G.; Dohn, A. O.; Moltke, A.; Vester, P.; Hansen, F. B. K.; Kjaer, K. S.; Hartsock, R.; Christensen, M.; Gaffney, K. J.; Henriksen, N. E.; Møller, K. B.; Haldrup, K.; Nielsen, M. M. Anisotropy enhanced X-ray scattering from solvated transition metal complexes. *Journal of Synchrotron Radiation* **2018**, *25*, 306–315, DOI: 10.1107/S1600577517016964.

(30) Dohn, A. O.; Jónsson, E. Ö.; **Levi**, G.; Mortensen, J. J.; Lopez-Acevedo, O.; Thygesen, K. S.; Jacobsen, K. W.; Ulstrup, J.; Henriksen, N. E.; Møller, K. B.; Jónsson, H. Grid-Based Projector Augmented Wave (GPAW) implementation of Quantum Mechanics/Molecular Mechanics (QM/MM) electrostatic embedding and application to a solvated diplatinum complex. *Journal of Chemical Theory and Computation* **2017**, *13*, 6010–6022, DOI: 10.1021/acs.jctc.7b00621.

(31) **Levi, G.**; Causà, M.; Lacovig, P.; Salatino, P.; Senneca, O. Mechanism and Thermochemistry of Coal Char Oxidation and Desorption of Surface Oxides. *Energy & Fuels* **2017**, *31*, 2308–2316, DOI: 10.1021/acs.energyfuels.6b02324.

(32) **Levi, G.**; Senneca, O.; Causà, M.; Salatino, P.; Lacovig, P.; Lizzit, S. Probing the chemical nature of surface oxides during coal char oxidation by high-resolution XPS. *Carbon* **2015**, *90*, 181–196, DOI: 10.1016/j.carbon.2015.04.003.